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Structural evolution and micromechanical properties of ternary Al–Ag–Ge alloy solidified under microgravity condition

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ABSTRACT

Microgravity solidification provides a particular opportunity to produce an extraordinary microstructure and optimized mechanical properties. To shed further light on the influence of rapid solidification mechanism on mechanical properties, both the microgravity solidification mechanism and resultant micromechanical properties of ternary $Al_{57}Ag_{12}Ge_{31}$ alloy were analyzed by means of drop tube, nanoindentation and frictional sliding techniques, which was compared with equilibrium solidification condition. The solidification pathways changed with the decrease of droplet size, owing to a larger cooling rate and a higher undercooling. Consequently, the microstructure transformed from dendrites plus two-phase eutectic to two-phase eutectics, eventually to anomalous ternary eutectic, while the thickness of surface (Ge) layer decreased. The micromechanical properties of rapidly solidified alloy droplets were evidently improved with the decrease of droplet size, which is mainly ascribed to the microstructure refinement and the homogenous distribution especially of hardening (Ge) phase. The measured microhardness, yield strength, strain hardening exponent, pile-up resistance and friction coefficient were analyzed as a function of droplet size.

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1. Introduction

The solidification behavior on the Earth is associated with gravity-dependent heat and mass transportation [1]. However, under the microgravity condition, the motion of liquid phase is impeded or altered by the interaction of buoyancy, reduced natural convection and thermocapillarity during solidification [2], a special microstructure may consequently form. The study of the solidification mechanism under this unique condition is significant for the investigation of the effect of natural convection and Marangoni motion. Due to the expensive costs and inconvenience of solidification experiments carried out on a space station or a space shuttle [3–5], researchers alternatively utilize a drop tube method to investigate the microgravity solidification phenomena in the metals and alloys. In the drop tube, metal or alloy melt is deeply undercooled to a metastable state and then rapidly solidified in a free fall at an acceleration of 9.8 ms^{-2} [6–9]. Consequently, the phase transitions may take place far away from an equilibrium state. Rapid solidification under microgravity provided by a drop tube method has the advantage of designing new materials with optimized mechanical properties.

There is thus a strong motivation for better understanding the relationship between microgravity solidification mechanism and the mechanical properties of the alloys. However, the problem is that the mechanical properties of the alloys prepared by a drop tube processing, i.e. the behaviors of stress–strain, tensile and friction, remain unclear owing to the measurement difficulty of solidified small-size droplets, although researches on the microgravity solidification and related microstructural formation has been carried out in the recent decades [6–11]. A nanoindentation technique has been successfully developed to conduct mechanical measurements for nano-/micro-sized complicated materials, e.g. complicated phases, thin films, metallic glasses, porous structures, biomaterials [12–16], and is thus employed in the present study to investigate micromechanical properties of alloy droplets solidified in a drop tube.

Al–Ag–Ge alloys, a member of ternary aluminum-based system, are nearly unwrapped material for either their solidification mechanism or mechanical properties [17,18]. The addition of Ag element will enhance the ductility of alloys while counteract the brittleness brought by Ge element. Ge as a semiconducting element

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is favored for electronics but does not attract much attention in an engineering field due to its extraordinary hardness and faceted crystal growth manner. The participation of Ge element brings a special growth relationship with other phases during solidification as well as hardens the materials [19,20], and the solidification of Al–Ag–Ge alloys may involve the formation of dendrite, two-phase eutectic and ternary eutectic. The solidification pathway influenced by solidification condition and processing parameters such as cooling rate, undercooling, etc., forthrightly determines the microstructural transition, subsequently changing the mechanical properties of the alloys, and needs to be clarified.

In this work, the microstructural characteristics of microgravity-solidified ternary $Al_{57}Ag_{12}Ge_{31}$ alloy droplets by means of a containerless drop tube process were hence investigated, and compared with the equilibrium solidification of the bulk alloy in a differential scanning calorimeter (DSC). The microgravity solidification pathways were proposed. The micromechanical properties were further analyzed by nanoindentation and friction sliding to reveal the influence of microgravity solidification mechanism on the consequent micromechanical properties of the ternary $Al_{57}Ag_{12}Ge_{31}$ alloy.

2. Experimental procedure

2.1. Rapid solidification at microgravity state

The master $Al_{57}Ag_{12}Ge_{31}$ alloy samples of 1 g for microgravity solidification were prepared from high-purity Al (99.999%), Ag (99.999%) and Ge (99.999%) in an arc melting furnace under an argon atmosphere. The microgravity solidification of the alloy was performed in a 3 m-long drop tube which was evacuated to a pressure of 2×10^{-4} Pa and refilled with the mixture gas composed of ultrapure argon and helium to about 1×10^5 Pa. The sample placed in a $\phi 16 \times 150$ mm quartz tube with a nozzle of $\phi 0.3$ mm size at its bottom, was melted and then superheated to about 200 K above their liquidus temperatures by induction heating for several seconds. The molten sample was ejected through the nozzle by high-pressure argon gas, and dispersed into fine liquid droplets which would be rapidly solidified with diameters of about 100–600 μm during free fall. In comparison, the equilibrium solidification of the alloy were accomplished by a differential scanning calorimeter (Netzsch DSC 404C, Germany) with a heating-cooling rate of 10 $Kmin^{-1}$, by which the liquidus temperature of the alloy was also measured. For the DSC experiment, the three pure metals (Al, Ag and Ge) of about 50 mg were heated together by laser melting (Trumpf HL 1006D Nd:YAG laser, Germany) with the protection of an argon flow for measurement accuracy.

2.2. Microstructure and composition characterizations

The solidified samples were polished and etched. Their phase constitution and microstructures were analyzed by an X-ray diffractometer (XRD, Rigaku D/max 2500 V, Japan), using Cu Kα radiation operated at a voltage of 40 kV, a current of 200 mA and a scanning rate of $5°min^{-1}$. The microstructures were observed using an optical microscope (OM, Zeiss Axiovert 200 MAT, Germany) and a scanning electron microscope (SEM, Tescan VEGA3 LMH, Czech) with energy dispersive spectrometry (EDS, Oxford INCA X-ACT, England).

2.3. Micromechanical property measurements

Microhardness and frictional sliding experiments were performed using an instrumented nanoindenter, (Nanotest™, Micromaterials, Wrexham, UK). A conical diamond tip (tip half angle of 70.3° and tip radius of 2 μm) was calibrated using a standard fused quartz sample. To obtain the average value of the whole sample and to avoid the effect of penetration depth as well as to consider the greatly different sizes of the droplets and the DSC sample, more than 200 large enough indentations with a load of 2 N (load-controlled; loading up in 20 s, holding for 10 s at the peak load and unloading in 20 s) were made on each sample. The microhardness and reduced elastic modulus values were calculated using the Oliver–Pharr method [21–23] according to load-displacement curves.

In a frictional sliding experiment, tangential loads generated during scratching were monitored by friction transducers mounted on either side of the indenter tip. After the tip was positioned at the marked location on the sample, the normal load was ramped up to a maximum value of 600 mN at a rate of 60 mNs^{-1}. With the such constant normal load, the tip was moved at a rate of 5 μms^{-1} throughout the cross-section of the droplets or the designed region of the DSC sample to form a permanent and steady scratch. The final surface profiles of the plastically deformed scratches (elastic deformation recovered) were obtained using a profilometer (Tencor P16, KLA-Tencor, USA) with a 2 μm radius diamond stylus. The pile-up heights and maximum depths of the scratches were determined from the surface profiles of the steady-state region of the sliding tracks.

3. Results and discussion

3.1. Equilibrium solidification during DSC analysis

3.1.1. Phase constitution

Fig. 1 shows the equilibrium phase diagram of Al–Ag–Ge ternary alloy system [24]. The composition of $Al_{57}Ag_{12}Ge_{31}$ (marked as point A) is located in the (Ge) phase region. The equilibrium solidification of the alloy was realized by a DSC experiment. Meanwhile, the liquidus temperature was also determined due to the lack of thermodynamic data for the Al–Ag–Ge alloy system. Fig. 2a plots the DSC heating–cooling curve. The liquidus temperature was determined to be 820 K, which is the melting temperature of primary (Ge) phase. 719 K and 691 K are the melting temperatures of two two-phase eutectics respectively, as discussed

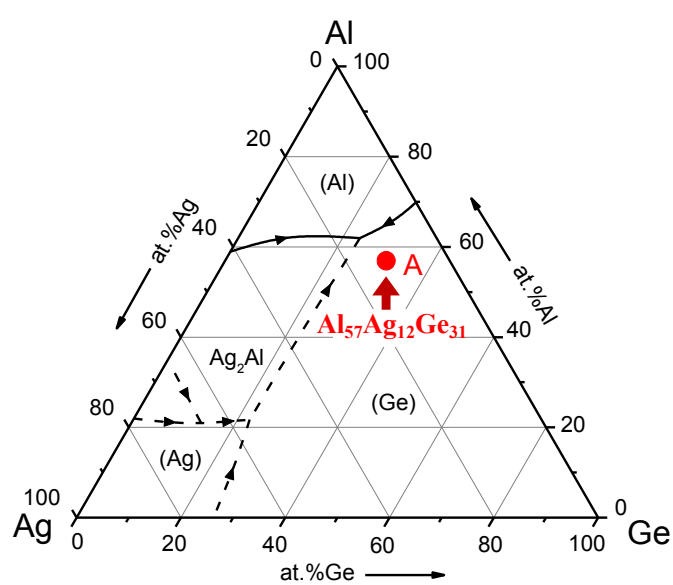

Fig. 1. $Al_{57}Ag_{12}Ge_{31}$ alloy (marked as point A) located in Al–Ag–Ge phase diagram [24].

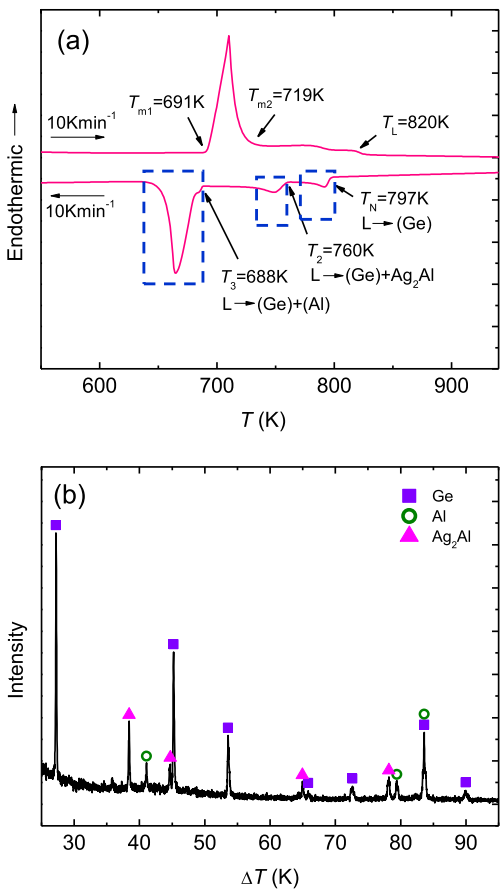

Fig. 2. Thermal analysis and phase constitution of $Al_{57}Ag_{12}Ge_{31}$ alloy: (a) DSC curve; (b) XRD pattern.

in the following section. The DSC sample was undercooled to 23 K. In the slow cooling process, three exothermic peaks with the onset temperatures of 688 K, 760 K and 797 K were detected, revealing that three phase transitions may take place during equilibrium solidification.

For the XRD analysis of phase constitution, an arc melted sample was prepared in a similar solidification due to the small mass of the DSC sample, as presented in Fig. 2b. The alloy is composed of (Ge), Ag_2Al and (Al) phases, which is in accordance with the information derived from Al–Ag–Ge phase diagram [24]. Both the strongest diffraction peak of (Ge) (111) at $2\theta = 27.22°$ and the largest peak sum of (Ge), compared with the diffraction peaks of the other two phases, imply that (Ge) may be the primary and dominant phase. According to the mass conservation law, theoretically, the fractions of (Ge), Ag_2Al and (Al) phases are 44.4, 28.3 and 27.3 wt%, consistent with the following microstructure analyses.

3.1.2. Microstructural formation mechanism

The three phases of equilibrium solidified DSC sample and their solute contents were further identified by OM, SEM and EDS analyses, as shown in Fig. 3 and Table 1. The solubilities of Ag and Al in the diamond-cubic faceted (Ge) solid solution are essentially zero, and their contents in the Ag_2Al intermetallic compound are 62.3 and 37.7 at.%, respectively, with no Ge. For the (Al) solid solution, the contents of Ge and Ag solutes increase to 2.7 and 2.8 at.% respectively, different from the situation in related binary alloy systems, i.e. the Ge solute content is zero in Al–Ge alloys at 273 K and Ag solute content is only 1 at.% in Al–Ag alloys at 473 K [25]. It suggests that both solid solubilities of Ge and Ag in the (Al) phase were slightly enlarged in the Al–Ag–Ge alloy in comparison with the related binary alloys.

The OM (Fig. 3a and b) and SEM images (Fig. 3c) show that the microstructure of DSC sample is composed of primary (Ge) block, Ag_2Al dendrite, (Ge) + Ag_2Al and (Ge) + (Al) two-phase eutectics. As determined by DSC traces (Fig. 2a), during solidification, the primary (Ge) phase was firstly generated from the melt at 797 K, corresponding to the first exothermic peak. Immediately, the phase transition L → (Ge) + Ag_2Al took place at 760 K, which is only 37 K behind. The (Ge) + Ag_2Al two-phase eutectic was then formed, companied with the slow growth of the primary (Ge) block simultaneously, as shown in Fig. 3a. For (Ge) + Ag_2Al two-phase eutectic, (Ge) phase still exhibited faceted characteristic owing to high melting enthalpy. At this stage, besides the formation of (Ge) + Ag_2Al two-phase eutectic, some Ag_2Al phase preferred to nucleate around the primary (Ge) block and grew independently to form dendrites (Fig. 3b). Therefore, both the Ag_2Al dendrite and the (Ge) + Ag_2Al two-phase eutectic are the solidification products for the second exothermic peak. At the last stage, (Ge) + (Al) two-phase eutectic was formed, instead of the ternary eutectic as suggested by the Al–Ag–Ge phase diagram [24]. The (Ge) + (Al) two-phase eutectic grew with a typical morphology observed in the Al–Ge binary alloys. Its growth size was refined because of the formation in the last stage solidification. According to the above analyses, the equilibrium solidification pathway is determined to be: L → (Ge) → (Ge) + Ag_2Al → (Ge) + (Al). It should be noted that both Ag_2Al dendrite and (Al) phase preferred to nucleate around the primary (Ge) block, and the primary (Ge) block served as heterogenous nuclei for them. The predominant formation of the primary (Ge) block resulted in the depletion of Ge solute around it.

3.2. Microgravity solidification of alloy droplets

3.2.1. Solidification pathways versus droplet size

Fig. 4 shows the morphologies of the whole alloy droplets according to droplet size *d*. The complex microstructures distribute homogenously inside the droplets, however an amount of (Ge) phase gathered on the surface to form a layer. The phase transformation pathways and microstructure transition of the alloy droplets with different sizes *d* are presented in Fig. 5. As droplet size decreased, the microstructure transformed from the mixture of primary (Ge) dendrite, fragmented Ag_2Al dendrite and regular (Ge) + (Al) two-phase eutectic to a granular (Ge) + Ag_2Al two-phase eutectic with an intergranular (Ge) + (Al) two-phase eutectic, and eventually to an anomalous (Ge) + Ag_2Al + (Al) ternary eutectic, as elucidated by the schematic solidification pathways in Fig. 6.

In the case of the droplets with $d \geq 400$ μm, for example, $d = 480$ μm (Fig. 5a), as temperature decreased, the primary (Ge) phase nucleated directly from the melt as a dendrite structure, different from the block structure formed in DSC sample. The branches of the primary (Ge) dendrite were comprised of (111) twins, demonstrated by our previous research on Ag–Cu–Ge alloy using an EBSD analysis [26]. As temperature further decreased, the composition of melt reached the L → (Ge) + Ag_2Al two-phase eutectic line. However the (Ge) + Ag_2Al eutectic could not be identified in the microstructure because the eutectic (Ge) phase preferred to nucleate and grow along the primary (Ge) dendrites serving as the heterogeneous nuclei [27,28]. Therefore, most of eutectic Ag_2Al phases nucleated around primary (Ge) dendrites, however grew independently in the form of fragmented dendrite. At the last solidification stage, the residual melt solidified into a regular (Ge) + (Al) two-phase eutectic structure when its composition was switched to the L → (Ge) + (Al) two-phase eutectic line. The solidification pathway in this situation is summarized as Path A

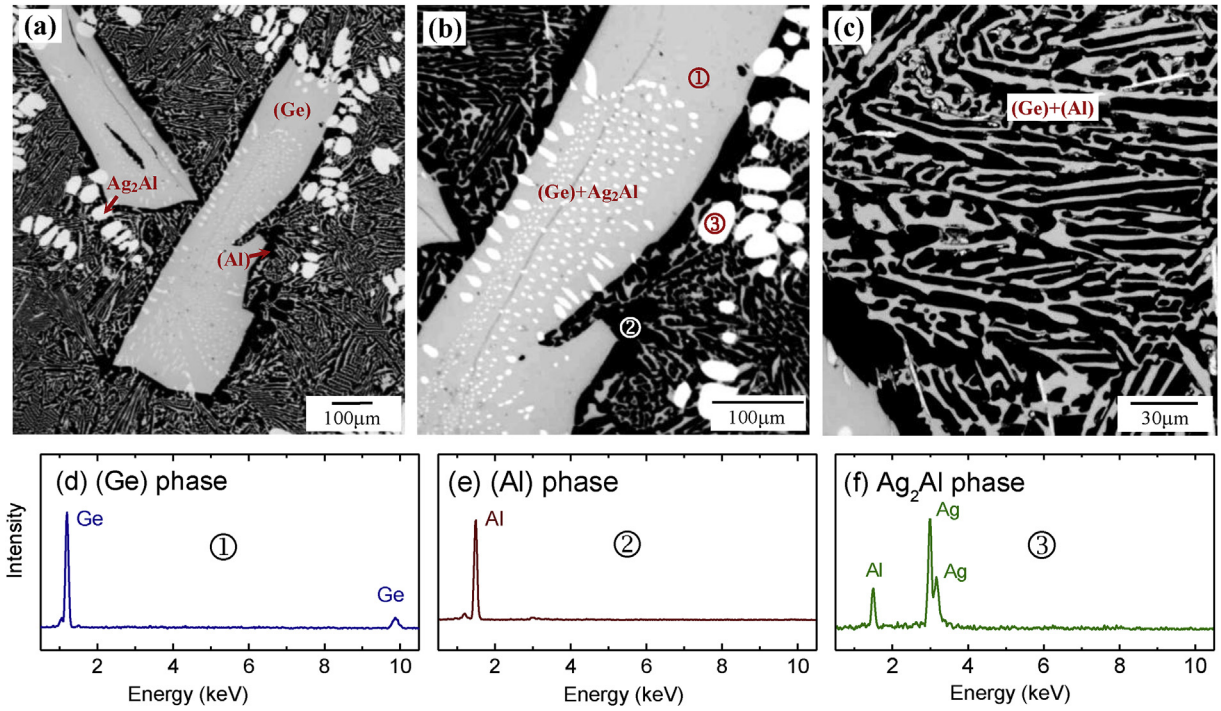

Fig. 3. Microstructures of DSC sample: (a) and (b) optical micrographs, (c) SEM micrograph, (b) and (c) are local enlargements of (a); (d)–(f) EDS analyses.

Table 1
Average solute contents of the three phases in DSC sample by EDS analysis.

Phase	Element (at.%)		
	Ge	Ag	Al
(Ge)	100	0	0
Ag_2Al	0	62.3	37.7
(Al)	2.7	2.8	94.5

in Fig. 6.

For 230 μm < d < 400 μm, the competition of both nucleation and growth between (Ge) and Ag_2Al phases contributed to a consequence that the two phases were generated from the melt simultaneously to form a granular (Ge) + Ag_2Al eutectic grain firstly, and soon after a regular (Ge) + (Al) two-phase eutectic was formed, as Path B illustrated in Fig. 6. In the case of the smallest droplets with $d \leq 230$ μm, much differently, the nucleation superiority of one or two leading phases was lost; consequently an anomalous ternary (Ge) + Ag_2Al + (Al) eutectic was rapidly formed, as Path C in Fig. 6, owing to the more drastic nucleation and growth competition among the three phases. From above, it is concluded that the microstructural formation mechanism is sensitive to droplet diameter under microgravity solidification condition. Actually, the variations of key solidification parameters, i.e. cooling rate and undercooling, with droplet diameter are believed to dominate the microstructural transition, as discussed below.

Due to the difficulty in the real-time measurements of cooling rate and undercooling of freely falling droplets in the drop tube, the cooling rate R_c with d were calculated using the Newtonian model regardless of the temperature gradient inside droplet as that [29].

$$R_c = \frac{6}{\rho C_p d}\left[\varepsilon\sigma\left(T^4 - T_0^4\right) + h(T - T_0)\right] \quad (1)$$

where ρ is the density of droplets, C_p the specific heat, ε the surface

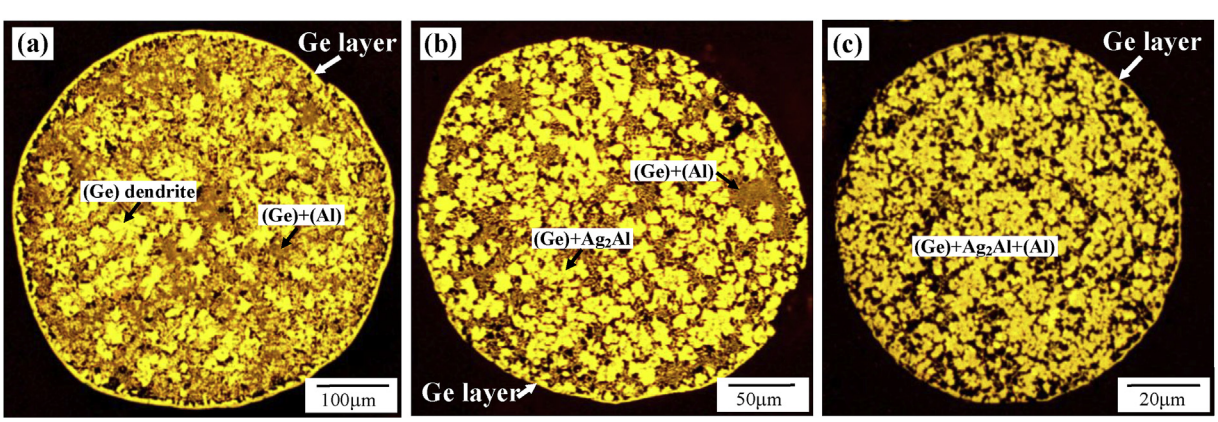

Fig. 4. Optical micrographs of alloy droplet profiles with different sizes: (a) d = 540 μm; (b) d = 300 μm; (c) d = 100 μm.

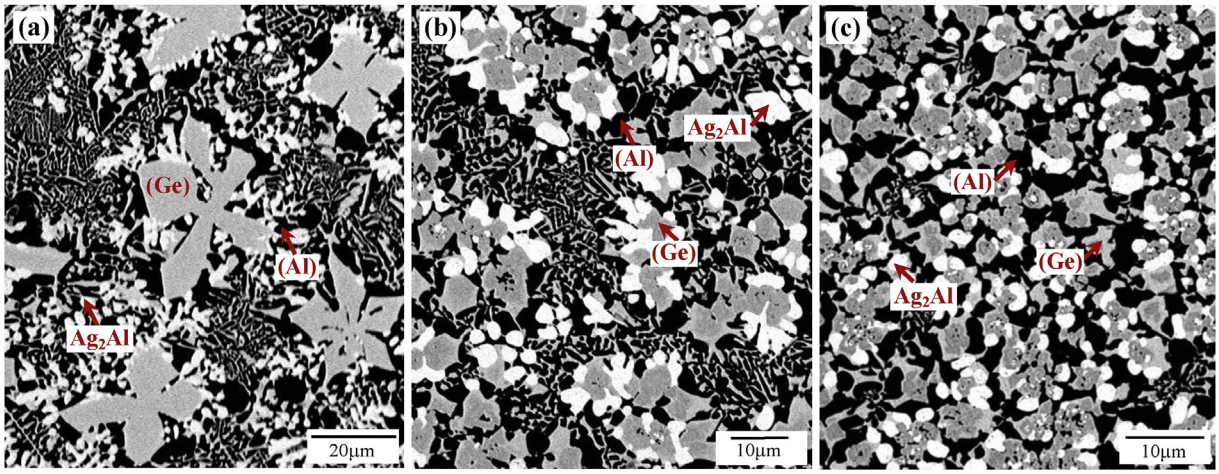

Fig. 5. SEM images showing microstructural morphologies of alloy droplets with different sizes: (a) $d = 480$ μm; (b) $d = 250$ μm; (c) $d = 140$ μm.

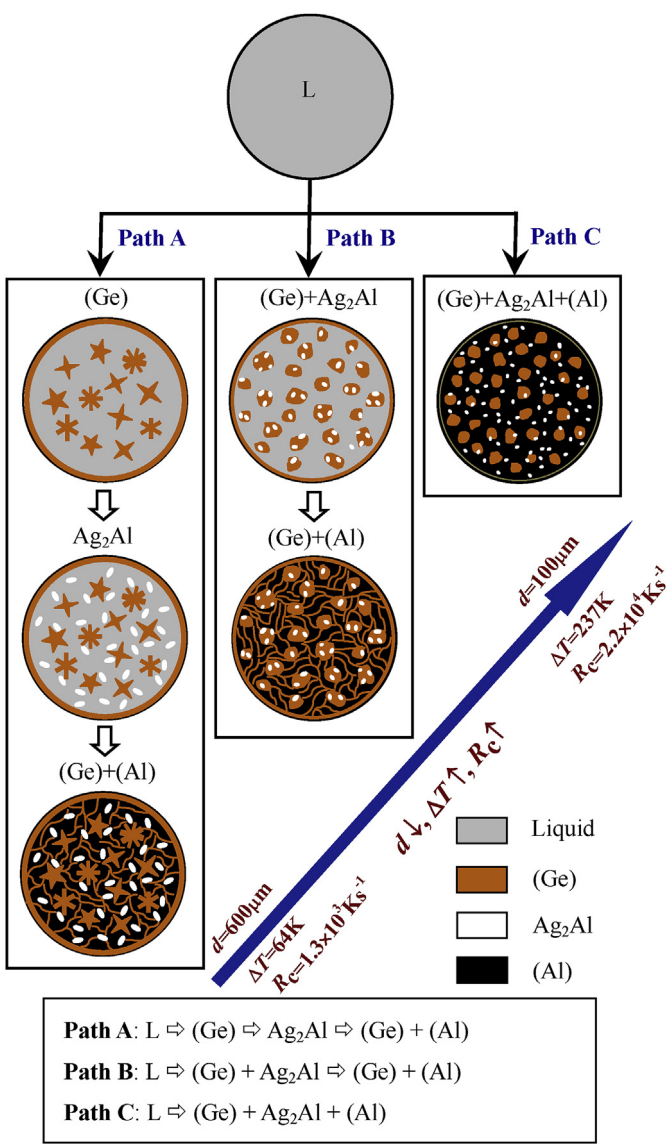

Fig. 6. Schematic illustrations of three microgravity solidification paths of alloy droplets.

emissivity, σ the Stefan–Boltzmann constant, T_0 the ambient temperature and h the heat transfer coefficient. Related parameters used in the calculations are listed in Table 2. Fig. 7a plots the temperature variations for the droplets with the diameter ranging from 100 to 600 μm. Moreover, the undercoolings ΔT of droplets were calculated according to a heat transfer model proposed by Lee et al. [32], as illustrated in Fig. 7b. Obviously, ΔT increases from 64 to 237 K ($0.29T_L$) as d decreases from 600 to 100 μm. According to the temperature variations and undercoolings, the initial cooling rates before falling R_{c1} and the cooling rate before solidification R_{c2} were also achieved, as given in Fig. 7b. With decreasing d, R_{c1} increased from 1.3×10^3 ($d = 600$ μm) to 2.2×10^4 Ks^{-1} ($d = 100$ μm), which yielded a higher ΔT following the relationship that

$$\Delta T = 36.3 + 2.1 \times 10^4 / d \tag{2}$$

3.2.2. Influence of microgravity on surface (Ge) layer formation

Phase separation phenomenon commonly occurs in some monotectic or peritectic alloys composed of the elements with a large difference in density [33–35], and possibly on the surface of these alloy droplets produced by gas atomization or drop tube processing [33,36]. On the surface of the microgravity solidified $Al_{57}Ag_{12}Ge_{31}$ alloy droplets rather than a DSC sample, a layer of (Ge) phase with a droplet size-dependent thickness was clearly observed to form, in consequence of the local fluctuation of solutal concentration insides the droplets, as discussed below.

Stokes motion, the natural convection motion driven by buoyancy caused from the difference in density among phases, influences the flow pattern of melt dominantly under the normal gravity condition. However, under a microgravity condition, the Stockes motion is restrained, while Marangoni convection motion (also called as thermo-capillary convection motion) takes the leading place. The Marangoni motion is driven by a surface tension gradient dependent on both temperature gradient and concentration gradient [37]. For an alloy droplet solidifying in the drop tube, as plotted in Fig. 8a, Ge-rich globules firstly departed from the melt and were ready for nucleation primarily into solid crystallites with the decrease of temperature. The movement of these Ge-rich globules was dominated by the coaction of the aforementioned three convections, i.e. the downward Stokes, the inward thermal Marangoni and the outward solutal Marangoni motions. The Ge-

Table 2
Physical parameters of $Al_{57}Ag_{12}Ge_{31}$ alloy.

Physical parameter	Value	Ref.
Liquidus temperature of alloy, T_L (K)	820	This work
Heat of fusion of alloy, ΔH_m (J kg^{-1})	3.69×10^5[a]	[30]
Density of liquid alloy, ρ (kg m^{-3})	4.44×10^3[a]	[30]
Specific heat of liquid alloy, C_p (J kg^{-1} K^{-1})	609[a]	[30]
Thermal conductivity of liquid alloy, k (W m^{-1} K^{-1})	101[a]	[30]
Surface emissivity of alloy, ε	0.21[a]	[30]
S/L interfacial energy of alloy, σ_{SL} (N m^{-1})	0.37	This work
Thermal conductivity of environmental gas, λ (J m^{-1} s^{-1} K^{-1})	1.8×10^{-2}	[31]
Temperature of environmental gas, T_g (K)	300	This work
Stefan-Boltzman constant, σ (W m^{-2} K^{-4})	5.67×10^{-8}	[30]
Boltzmann constant, k_B (J K^{-1})	1.38×10^{-23}	[30]

[a] Calculated according to the atomic fractions of pure elements [30].

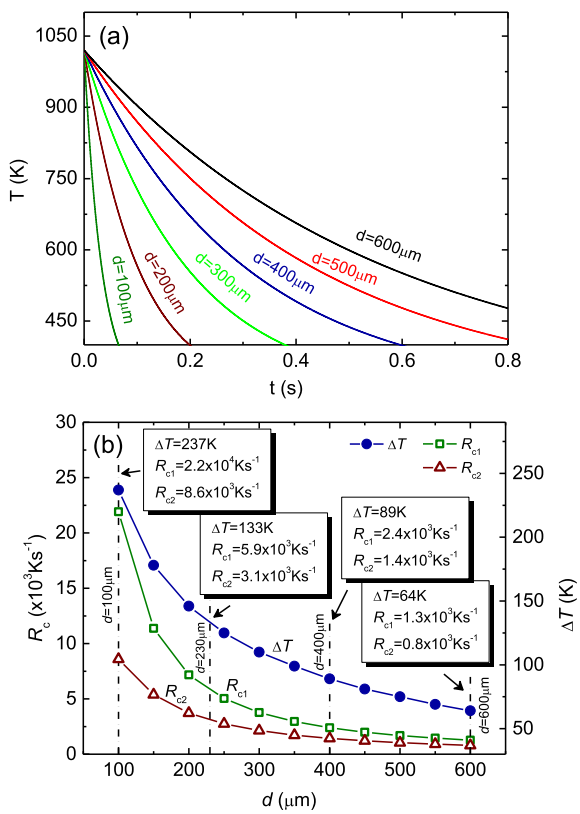

Fig. 7. Calculated temperature profile and cooling rate of alloy droplets: (a) cooling curves; (b) initial cooling rate before falling R_{c1}, cooling rate before solidification R_{c2} (at the moment before nucleation) and undercooling ΔT for the alloy droplets.

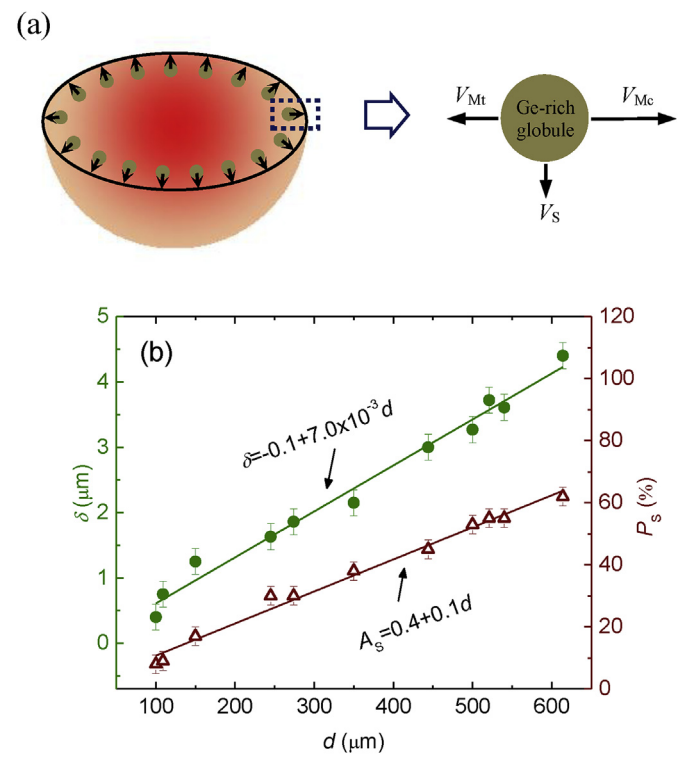

Fig. 8. Formation of surface (Ge) layer: (a) schematic formation process of surface (Ge) layer on rapidly solidified alloy droplets; (b) thickness δ and occurrence probability P_S of surface (Ge) layer versus droplet diameter d.

rich globules far away from the droplet surface nucleated quickly as the primary solid (Ge) phase observed in Fig. 4. However, at the surface with relative smaller surface tension under the effect of surface segregation, as demonstrated by the phase field simulations of surface layer that specially took surface free energy into account [33,36,38], the solute concentration fluctuated much drastically, i.e. with a large concentration gradient ∇C. Consequently, the solutal Marangoni provide a high fluid velocity V_{Mc} for mass transport; the closer to the surface, the greater the V_{Mc}. Owing to the superiority of V_{Mc} over the fluid velocity V_{Mt} caused by thermal Marangoni motion, the Ge-rich globules near the droplet surface would further move outwards, yielding the formation of surface (Ge) layer as also seen in Fig. 4.

Obviously, the thickness δ (or the occurrence probability P_S) of surface (Ge) layer varies, linearly, with droplet diameter d, as illustrated in Fig. 8b, following

$$\delta = -0.1 + 7.2 \times 10^{-3} d \quad (3)$$

$$P_S = 0.4 + 0.1d \quad (4)$$

It suggests that, as d decreases, the (Ge) phase hardly form a surface layer but distribute more homogenously in the droplet. As acquired above that $\Delta T \propto 1/d$ and $R_{c2} \propto 1/d^2$, accordingly $\delta \propto 1/\Delta T$ and $\propto 1/R_{c2}^{1/2}$ are expected. Two reasons are believed to be responsible for the reduced δ in small droplets: (i) Thermodynamically, a deep undercooling ΔT yields greatly large driving force for the liquid-solid phase transformation, therefore the number of embryos and the rate of homogeneous nucleation of primary (Ge) phase in the droplets will markedly increase; (ii) Kinetically, a high cooling rate R_{C2} diminishes the duration of solidification, so there will not be sufficient time for the surface segregation of (Ge)

globules from the melt or the growth of surface (Ge) layer, hence yielding a thinner surface (Ge) layer as shown in Fig. 8b.

3.3. Micromechanical behaviors versus microstructural transition

3.3.1. Microhardness of rapidly solidified alloy droplets

Similarly, influenced by the above two factors, ΔT and R_{c2}, the microstructure of microgravity solidified alloy droplets change, as observed in Fig. 4, leading to the variation of measured microhardness H_m of the droplets with different diameters, as given in Fig. 9. With the decrease of d and the increase of ΔT and R_{c2}, the temperature gradient between the surface and the center of the droplets was lowered, and the expelling of solutes (to achieve constitutional undercooling) during the solidification of primary (Ge) phase was suppressed, impeding the growth of dendrite structure but facilitating more homogeneous nucleation of the (Ge) phase. The droplets were hence markedly hardened in consequence of obvious microstructure refinement, i.e. H increased from 2.2 (d = 600 μm, ΔT = 64 K, R_{c2} = 0.8 × 10^3 Ks^{-1}) to 3.9 GPa (d = 120 μm, ΔT = 211 K, R_{c2} = 7.2 × 10^3 Ks^{-1}).

The H_m-d relationship in Fig. 9 accords with the microstructural transition of the droplets and is categorized into three regimes with d. As summarized in Section 3.2.2, the microstructure of Regime 1 consists of: primary (Ge) dendrite, Ag_2Al dendrite and (Ge) + (Al) two-phase eutectic (simplified as dendrites plus binary eutectic); Regime 2: (Ge) + Ag_2Al and (Ge) + (Al) two-phase eutectics (binary eutectics); Regime 3: anomalous (Ge) + Ag_2Al + (Al) ternary eutectic (ternary eutectic). Correspondingly, with the decreasing d, H increased slowly in Regime 1 but comparatively rapidly in Regimes 2 and 3 as the coarse dendrite structures clearly transformed into fine eutectics. It implies that, undoubtedly, the (Ge) phase acts as the hardening phase for the alloy, and the microstructure refinement and homogenous distribution especially of the (Ge) phase more dominantly improve the mechanical properties of the alloy droplets.

Because the microstructure of DSC sample is much coarser than that of a droplet, individual microhardness measurements on its Ge-rich and eutectic-rich regions were carried for comparison, utilizing the same testing parameters (Section 2.3). As expected, the microhardness H_m of the Ge-rich region is 6.53 GPa, approximately four times the value of the eutectic-rich as only 1.49 GPa. To better understand the overall mechanical performance of the whole DSC sample, the Halpin-Tsai equation was used to predict the average microhardness [39,40]:

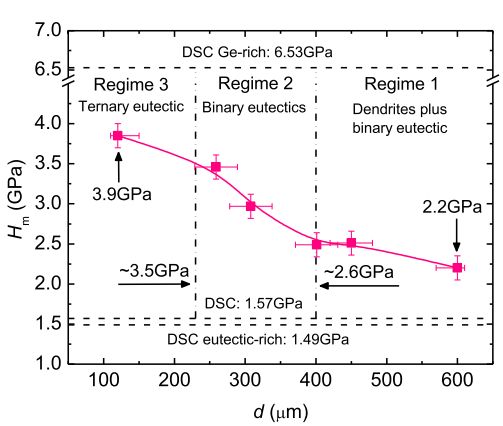

Fig. 9. Measured microhardness H_m of rapid solidified alloy droplets versus droplet diameter d.

$$\frac{P}{P_M} = \frac{1 + \xi\phi V_R}{1 - \phi V_R} \tag{5}$$

where P is the load or stress, the subscripts R and M the reinforcement and the matrix, respectively, V_R the volume fraction of reinforcement, ξ the measure of the reinforcement that depends on boundary conditions, $\phi = \frac{E_R/E_M - 1}{E_R/E_M + \xi}$, and E the modulus. We assumed the primary (Ge) dendrite as the reinforcing phase in the matrix of mixed Ag_2Al dendrite and (Al) + (Ge) eutectic for the DSC sample (E_{Ge} = 103 GPa, E_{Ag2Al} = 79 GPa and E_{Al} = 70 GPa [41]). Approximately, $P/P_M = [(P_{11}/P_M + 2P_{22}/P_M)]/3$ [39] (P_{11} and P_{22}: the loads or stresses in longitudinal and transverse directions respectively) was calculated to be 1.0557, and the average microhardness of the whole DSC sample is estimated to be only 1.57 GPa, much lower than the value of alloy droplets due to its coarse microstructure.

3.3.2. Yield, hardening and friction performances

For elastic-plastic materials, pile-up or sinking-in occurs during frictional sliding. Information on pile-up or sinking-in is constructive for both comprehensively understanding contact mechanics and estimating other micromechanical properties, i.e. yield strength σ_y and strain hardening exponent n. Fig. 10a is the 3D surface topography image of a scratch on a solidified alloy droplet with d = 530 μm in the frictional sliding experiment. The pile-up phenomenon on both sides of the scratch groove is evidently visible, and the cross-section profile of this scratch is plotted in Fig. 10b (h_p: pile-up height, h_r: residual penetration depth, x: scanning distance). The values of h_p and h_r are presented in Fig. 10c, as linear functions of d that

$$h_p = 1.5 + 2.0 \times 10^{-3} d \tag{6}$$

$$h_r = 4.6 + 1.1 \times 10^{-3} d \tag{7}$$

In comparison with the results of DSC sample (h_p = 4.12 μm and h_r = 5.95 μm for a eutectic-rich region, h_p = 2.31 μm and h_r = 4.37 μm for a Ge-rich region), h_r of the droplets was between the values of the Ge-rich region and the eutectic region, while h_p of the droplets was around or even below the value of the Ge-rich region, both being related to the hardness of the samples. The normalized pile-up height r_p ($r_p = h_p/h_r$) was further calculated, as plotted in Fig. 10d, and a linear relationship was obtained as that

$$r_p = 0.3 + 2.8 \times 10^{-4} d \tag{8}$$

It was interestingly found that, even the droplets were softer than the Ge-rich region of the DSC sample, however, all the r_p values of the droplets were lower than that of the Ge-rich region, revealing the different yielding and strain hardening behaviors of the solidified droplets.

The pile-up behavior is a function of σ_s/E^*(σ_s: initial yield strength, E^*: reduced modulus) and strain hardening exponent n [42,43]. By using the chart proposed by Bellemare et al. [44], with logarithmic σ_s/E^* as the ordinate and n as the abscissa, σ_s and n can be estimated from the scratch microhardness and the pile-up height (E^* measured as 86 GPa for the droplets), as presented in Fig. 10e. As d decreased from 600 to 120 μm, apparently, σ_s increased from 0.32 to 0.45 GPa, similar to the trend of microhardness (Fig. 9), in accordance with the microstructural transition of the droplets. The relationship between σ_s and H is plotted in Fig. 10f, as that

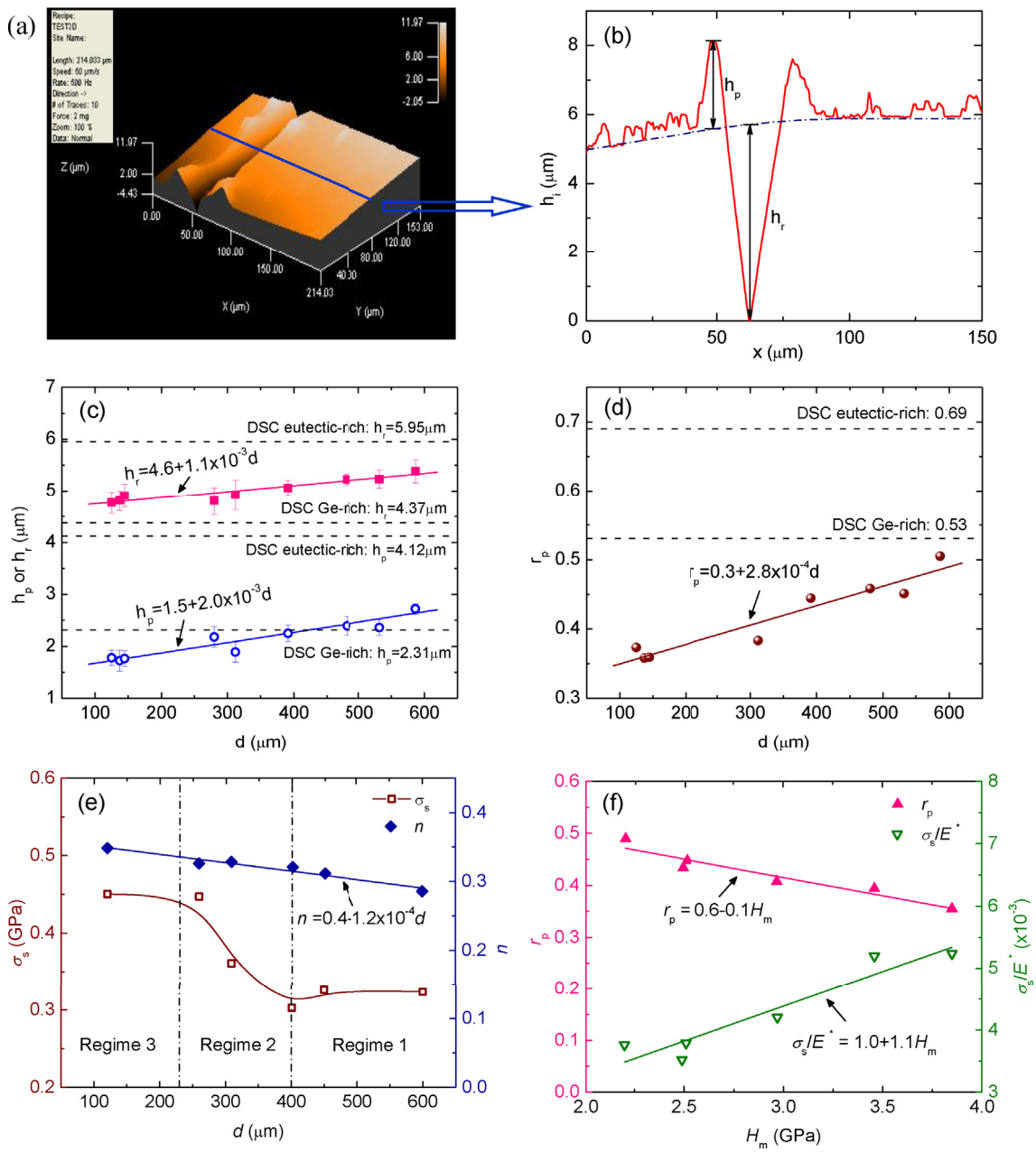

Fig. 10. Micromechanical properties of rapidly solidified alloy droplets: (a) 3D topography image of a scratch for a 530 μm droplet; (b) cross-section profile of the scratch in (a) with the definitions of pile-up height h_p and residual penetration depth h_r; (c) h_p and h_r versus droplet size d; (d) normalized pile-up height r_p versus d; (e) yield strength σ_s and strain hardening exponent n versus d; (f) r_p and σ_s/E^* versus microhardness H_m.

$$\sigma_s/E^* = 1.0 + 1.1H_m \tag{9}$$

n slightly increased with the decrease of d, from 0.29 to 0.35, following the linear equation (as shown in Fig. 10e)

$$n = 0.4 - 1.2 \times 10^{-4}d \tag{10}$$

Based on the difficulty in yielding and the facility for strain hardening, the pile-up behavior of the droplets with a small diameter (i.e. a high microhardness) is minimized, following the linear relationship

$$r_p = 0.6 - 0.1H_m \tag{11}$$

For aluminum alloys, wear behavior is another crucial mechanical property attracting more attention, which is basically determined by the hardness and the friction coefficient of materials. Hence, the friction coefficients f between the indenter tip and the microgravity solidified alloy droplets along scratch direction are acquired and plotted in Fig. 11a, compared with DSC sample. For the non-uniform DSC sample, f fluctuated much greater in the Ge-rich region than in the eutectic-rich region owing to the fracture of brittle Ge. For the droplets, the fluctuation of f lessened with decreasing d, indicating a more homogeneous and refined microstructure of the smaller droplet. As given in Fig. 11b, with the decrease of d, the average friction coefficient f_m decreased from 0.40 (d = 600 μm) to 0.35 (d = 120 μm) with a linear relationship

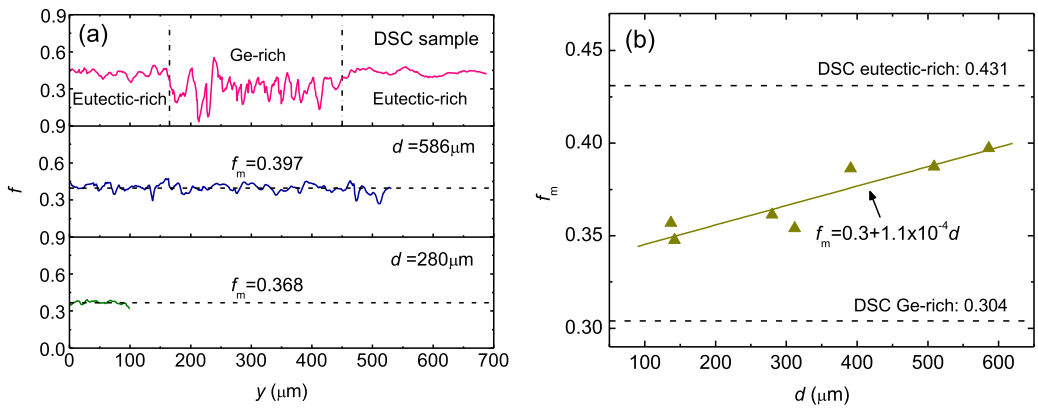

Fig. 11. Friction behaviors of rapidly solidified alloy droplets compared with DSC sample: (a) friction coefficient f along scratch direction y; (b) average friction coefficient f_m versus d.

$$f_m = 0.3 + 1.1 \times 10^{-4} d \quad (12)$$

Associated with a higher microhardness, the smaller friction coefficient of smaller droplets is anticipated to yield higher wear resistance for a potential tribological application.

4. Conclusions

In summary, the microgravity solidification of ternary $Al_{57}Ag_{12}Ge_{31}$ alloy droplets was investigated using a drop tube. Their micromechanical properties were analyzed and compared with those under equilibrium solidification condition. As droplet diameter decreased from 600 to 120 μm, their solidification pathways changed, owing to an enlarged cooling rate to 7.2×10^3 Ks^{-1} and a deepened undercooling to 211 K. Accordingly, the microstructure of the droplets transformed from the mixture of primary (Ge) dendrite, fragmented Ag_2Al dendrite and regular two-phase eutectic composed of (Ge) and (Al) phases, to a granular two-phase eutectic composed of (Ge) and Ag_2Al phases with an intergranular two-phase eutectic composed of (Ge) and (Al) phases, and eventually to an anomalous ternary eutectic. The thickness of surface (Ge) layer caused by Marangoni motion decreased, and the distribution of (Ge) phase became more homogenous. The microstructure refinement and the homogenous distribution especially of the hardening (Ge) phase contributed to the increases of microhardness from 2.2 to 3.9 GPa and yield strength from 0.32 to 0.45 GPa, and their increasing trends classified into three regimes were in accord with the microstructural transition. Accordingly, the strain hardening exponent increased from 0.29 to 0.35, while the normalized pile-up height decreased from 0.51 to 0.37 and friction coefficient from 0.40 to 0.35, showing linear correlations with the droplet size.

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